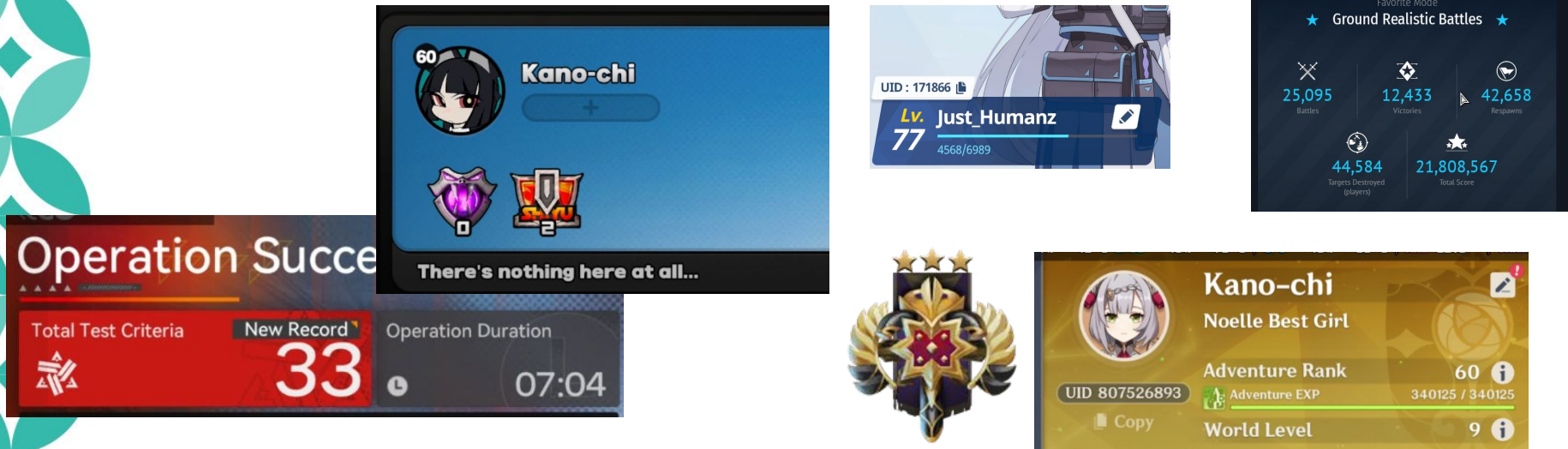


Gaming with CNCF Infrastructure

Aldin Setiawan

Quick Intro About me

A linux ~~gacha-addiction~~ gamer since 2016 and currently working as linux infrastructure stuff.



Throwback: Gaming on linux?

- Painful
- Unplayable
- Wasting time
- Can It run crysis?



Throwback: How hard it's is?

- Only a few of game are native running on linux(Dota2&CS:GO)
- Garbage desktop driver(Nvidia)
- Poor performance of OpenGL
- Tuning the kernel for low latency support
- Random crash



Nowadays: How hard it is?

- GGEZ
- Optimum Driver support
- ‘Just work’
- Most of dependency solved by containerized(flatpak)



Timeline of Gaming on Linux

- 2013 – SteamOS alchemist Release
- 2016 – Vulkan Project are Started
- 2017 – Wine Dxvk Introduce
- 2018 – Steam Proton Release
- 2021 – Steam Deck Booming

Gaming with CNCF Infrastructure?

Gaming on Linux now days are pretty easy(just work). Can we bring it into little bit further?

Like. Gaming on the Cloud?



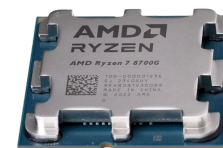
The Infrastructure

MiniPC as Server:

- AMD Ryzen 5 2400GE
- AMD ATI Radeon Vega Series
- 32GiB of Ram DDR4
- 2 NIC for 802.3ad Bond

PC as the Game Host:

- AMD Ryzen 7 8700G
- NVIDIA GeForce RTX 3090
- 16GiB of Ram DDR5
- 2 NIC for 802.3ad Bond

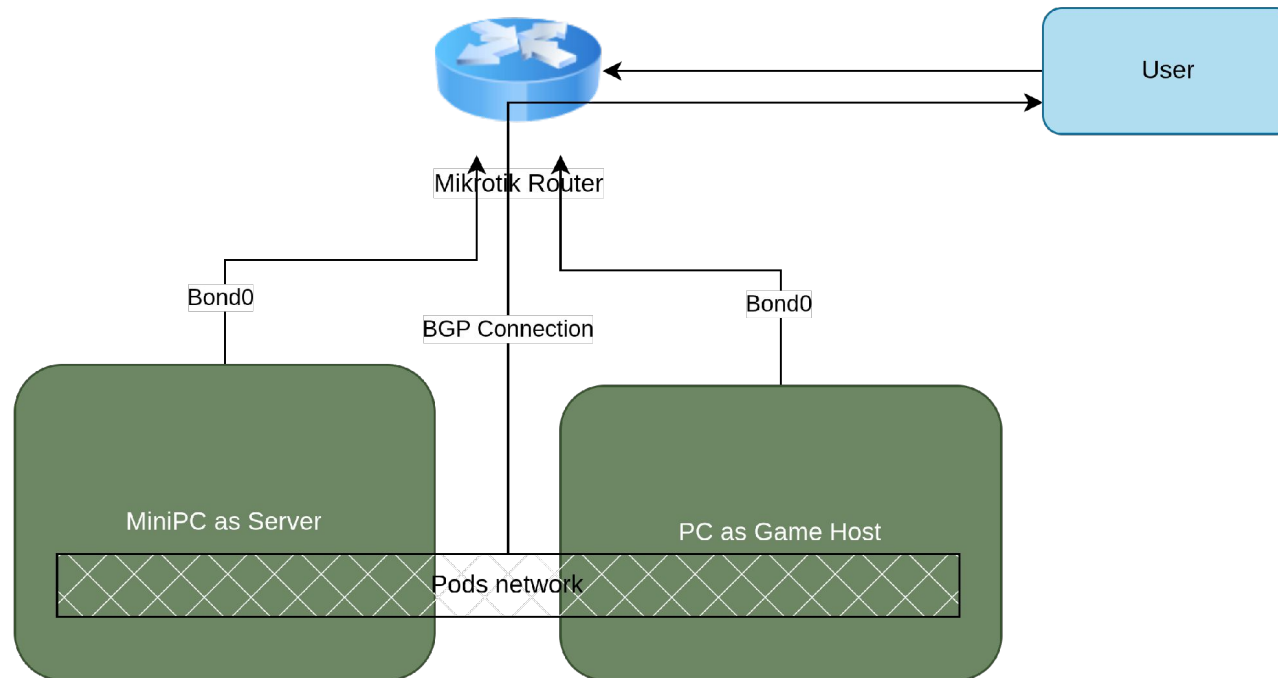


Mikrotik Router:

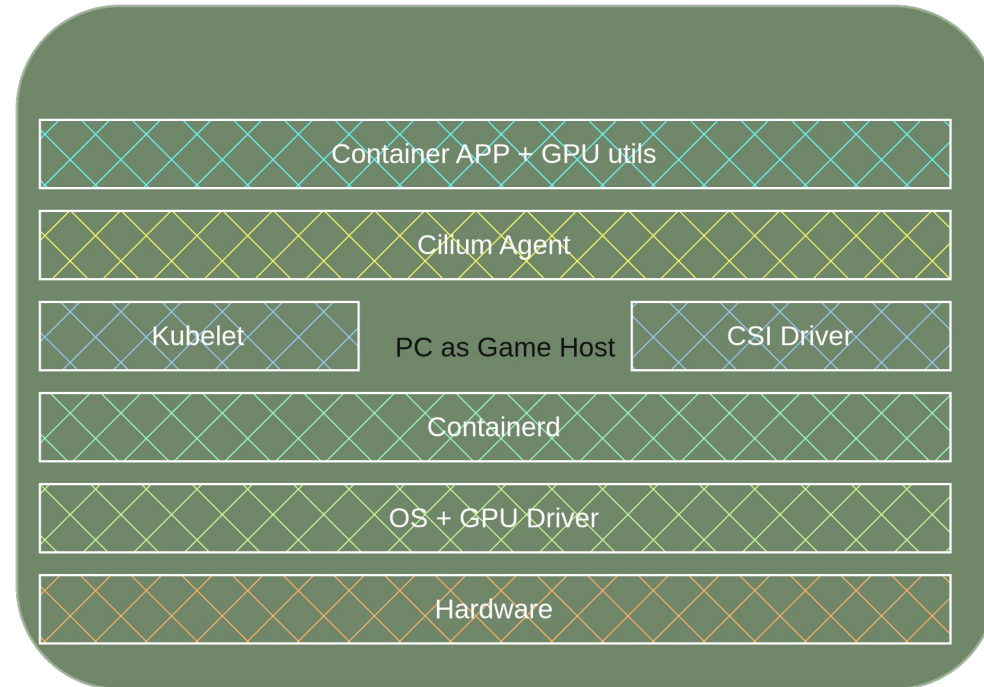
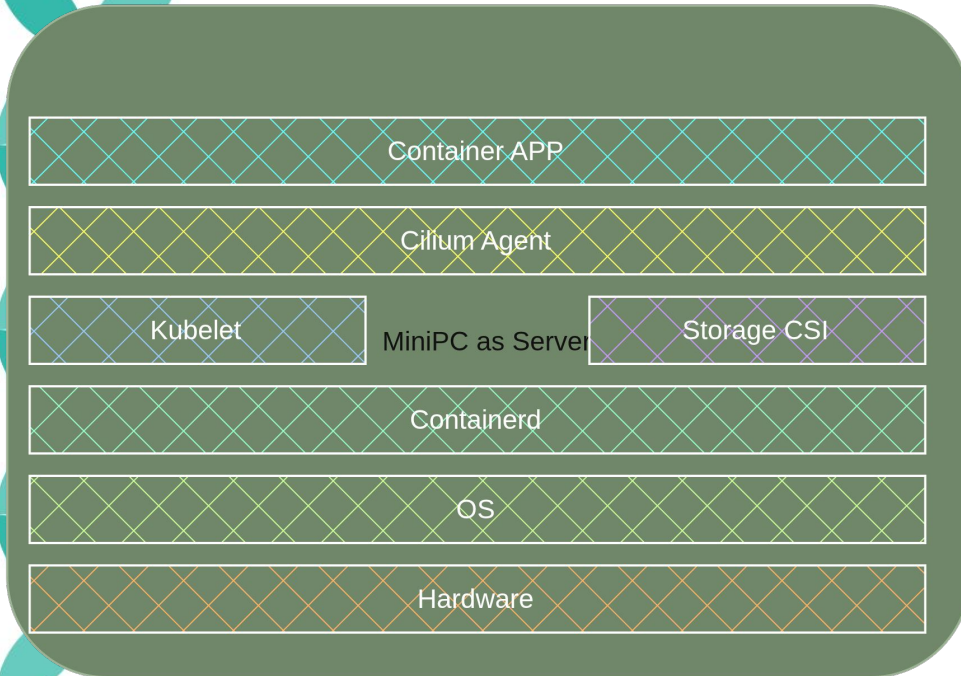
- L009UiGS-2HaxD
- 512MiB of Ram
- 4 NIC for 802.3ad Bond



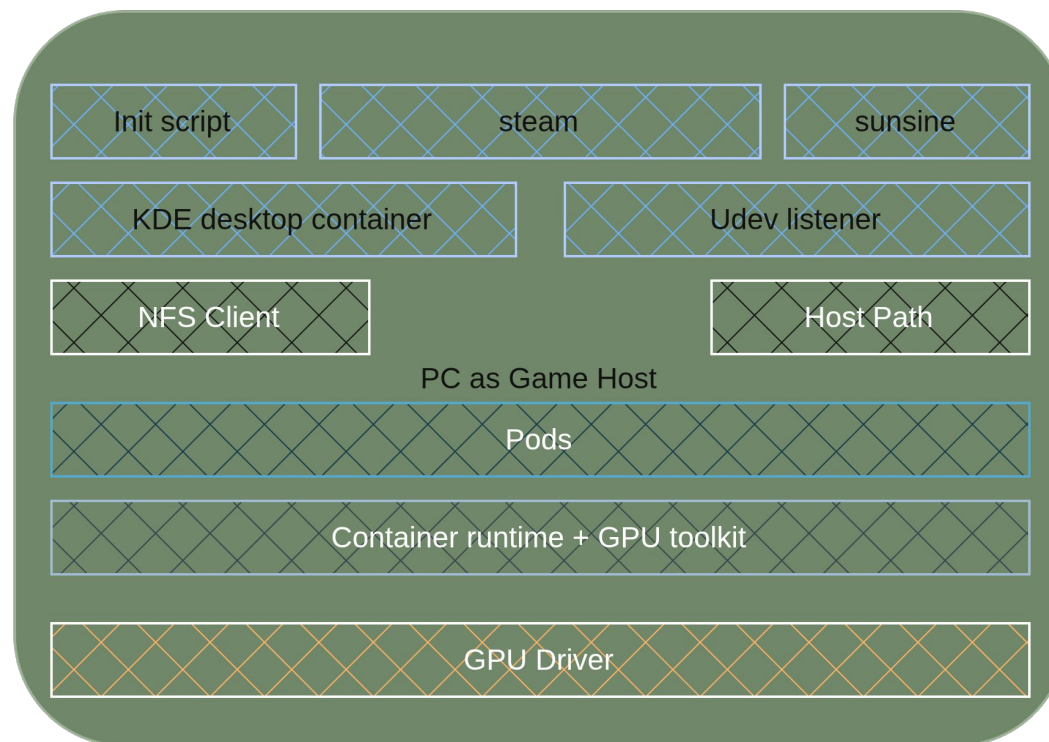
High Level design



High Level design



High Level design



Tools

- Rook
 - Ceph-rbd from rook
- Cilium
 - Pods Network
 - Pods BGP
- CoreDNS
 - Resolver for the stateless pods
- Helm
- Grafana+Prometheus
 - Monitoring Host&CSI

Quick Code Peek

Helm charts:

- [Github](#)

Quick Demo

[PC Desktop](#)

[Smartphone](#)

Is it done, Yuri?



PoC or Prototype

Huge blocker

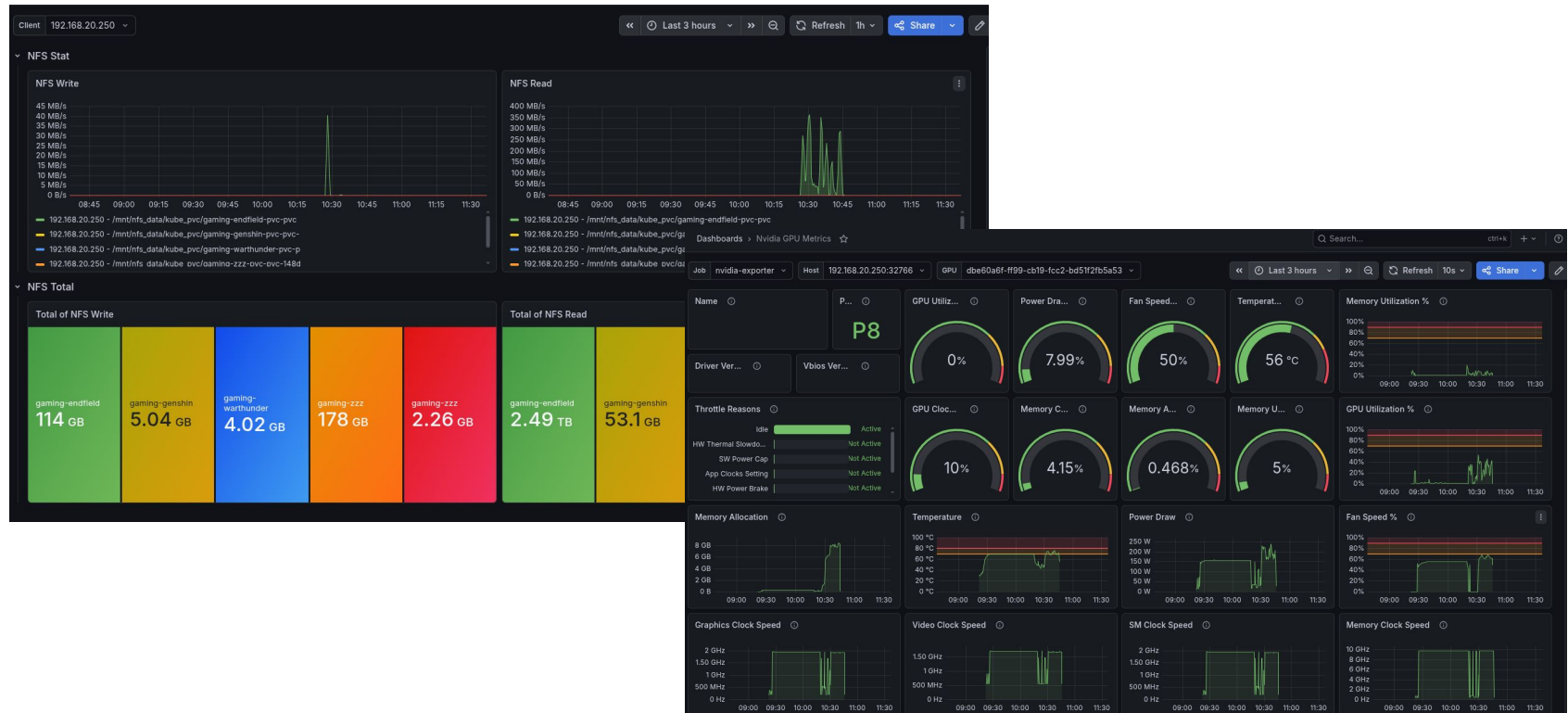
- Not all game support proton
- Competitive game not very recommended
- Anti Cheat ~~Spyware~~ Program
- End-User License Agreement

Point Plus

- Centralized Storage
- Immutable user-spaces
- Play anytime&anywhere



Monitoring (Optional)





Thank You



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